Infron vs OpenRouter Routing, Cache, Cost, and Streaming Performance A/B Test Report

Abstract and Executive Outline

Keywords: Prompt Caching; A/B Testing; Provider Routing; Cache Affinity; Latency; Throughput; Cost Attribution; qwen3.7-flash

Abstract

This report evaluates `qwen/qwen3.7-flash` on Infron and OpenRouter across cache reuse, observed cost, throughput, E2E latency, and Streaming TTFT under Prompt Caching workloads.

The main findings are: Infron leads Cache hit rate in 1/4 routing modes; OpenRouter leads in 3/4; 0/4 tie; Infron leads Observed cost in all routing modes; Infron leads Throughput in all routing modes; OpenRouter leads E2E latency in all routing modes; Infron leads Streaming TTFT in 1/4 routing modes; OpenRouter leads in 3/4; 0/4 tie.

Overall, Infron's cross-mode strengths are throughput, observed cost, while OpenRouter's cross-mode strengths are Streaming TTFT, E2E latency, cache reuse. Platform choice should be driven by workload objective rather than a single headline metric.

Figure 0: Normalized Capability Radar

The five radar axes represent throughput, price, E2E latency, Streaming TTFT, and cache hit rate. Every metric is normalized to a 0–100 score, with farther outward meaning better.

Thick solid lines show platform-level contours; translucent lines and points show individual routing modes.

Conclusion Overview: Core Metric Winners by Routing Mode

Based on strict A/B paired samples. Blue represents Infron, orange represents OpenRouter; gold cells mark the winner for the routing objective.

Throughput	Observed cost	E2E latency	Streaming TTFT	Cache hit rate
Infron wins 4/4 Max advantage 12.24%, higher is better	Infron wins 4/4 Max advantage 75.91%, lower is better	OpenRouter wins 4/4 Max advantage 97.35%, lower is better	OpenRouter wins 3/4 Max advantage 4.02%, lower is better	OpenRouter wins 3/4 Max advantage 25.07%, higher is better

Routing mode	Throughput objective	Cost objective	Latency objective	TTFT objective	Cache result
Throughput First throughput	Infron advantage 10.97%	Infron advantage 37.92%	OpenRouter advantage 95.81%	OpenRouter advantage 4.02%	OpenRouter advantage 25.07%
Price First price	Infron advantage 12.24%	Infron advantage 67.92%	OpenRouter advantage 82.73%	Infron advantage 1.19%	OpenRouter advantage 3.92%
Latency First latency	Infron advantage 11.57%	Infron advantage 68.27%	OpenRouter advantage 97.35%	OpenRouter advantage 3.40%	Infron advantage 0.23%
TTFT First ttft	Infron advantage 10.93%	Infron advantage 75.91%	OpenRouter advantage 96.15%	OpenRouter advantage 1.43%	OpenRouter advantage 4.88%

Executive Outline

Dimension	Conclusion	Evidence
Controls	First/second `usage.prompt_tokens` deltas are limited to 50 tokens within each `sort/group/round` pair.	Methods and data quality
Cache reuse	Infron leads Cache hit rate in 1/4 routing modes; OpenRouter leads in 3/4; 0/4 tie	Overall metrics and mechanism section
Observed cost	Infron leads Observed cost in all routing modes	Overall metrics and provider drill-down
Performance	Infron leads Throughput in all routing modes; OpenRouter leads E2E latency in all routing modes; Infron leads Streaming TTFT in 1/4 routing modes; OpenRouter leads in 3/4; 0/4 tie	Charts and statistical tests
Attribution boundary	Claims use observable response telemetry: usage, cost, TTFT, latency, provider fields, and cache tokens.	Provider/Route drill-down
Business meaning	Long-context, RAG-prefix, agent-tool, and high-frequency template workloads should evaluate cache rate, cost, first-token latency, and E2E latency together.	Discussion and conclusion

Routing-Mode Conclusions

Routing mode	Objective	Cache winner	Cost winner	Throughput winner	E2E latency winner	TTFT winner	Interpretation
Throughput First	Maximize output capacity per unit time	**OpenRouter**	**Infron**	**Infron**	**OpenRouter**	**OpenRouter**	OpenRouter leads overall (3/5 metrics)
Price First	Minimize request and token cost	**OpenRouter**	**Infron**	**Infron**	**OpenRouter**	**Infron**	Infron leads overall (3/5 metrics)
Latency First	Minimize full-response waiting time	**Infron**	**Infron**	**Infron**	**OpenRouter**	**OpenRouter**	Infron leads overall (3/5 metrics)
TTFT First	Minimize streaming first-token time	**OpenRouter**	**Infron**	**Infron**	**OpenRouter**	**OpenRouter**	OpenRouter leads overall (3/5 metrics)

1. Introduction: Background, Questions, and Contributions

LLM inference-platform behavior is shaped by provider routing, prompt caching, streaming response handling, cost attribution, and fallback policy. This report treats the platform as an observable system and measures speed, cost, cache reuse, and first-token experience through strict A/B pairs.

1.1 Research Hypotheses

Hypothesis	Statement	Validation metric
H1	Stronger provider/cache affinity improves token-level cache hit rate for repeated stable prefixes.	Second-request cache-read tokens and token cache hit rate
H2	Higher cache hit rate can reduce observed cost, but does not necessarily reduce TTFT or E2E latency.	Observed cost, average TTFT, average request latency
H3	Different routing sorts change provider selection and produce different cost/throughput/latency frontiers.	Provider distribution, throughput, latency, cost

1.2 Contributions

- Uses response-returned `usage.prompt_tokens` as the input-token control while allowing small cross-platform accounting variance up to 50 tokens.
- Extends prompt-caching evaluation to cost, throughput, E2E latency, TTFT, provider distribution, reasoning telemetry, and paired statistical tests.
- Scopes every claim to observable response telemetry instead of private routing internals.

2. Experimental Design, Dataset, and Controls

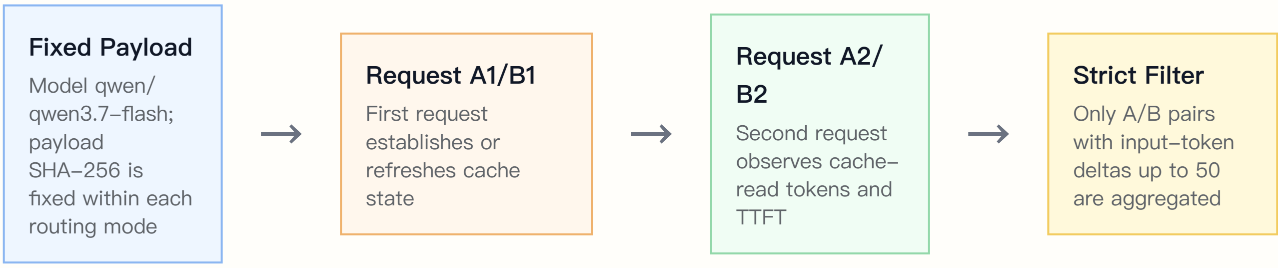
2.1 Dataset Construction

The dataset is `controlled_cache_probe`, covering 4 routing sorts, 2 platforms, 4 groups, and 50 rounds per group. Each round sends identical first/second Chat Completions requests; the second request observes cache-read tokens, TTFT, and E2E latency.

The built-in business templates represent stable long-context workloads such as RAG support, agent tool instructions, marketing automation, and code review.

2.2 Controlled Variables

Figure 1: Experimental Design and Strict A/B Pairing Filter



Controlled-variable rule: within the same `sort/group/round`, both platforms must have first/second `usage.prompt_tokens` deltas no greater than 50 tokens. Total Input Tokens are computed from response-returned usage, not local tokenizer estimates.

2.3 Metric Definitions

Metric	Definition	Direction
Call cache hit rate	Share of second requests with `cache_read_tokens > 0`	Higher is better
Token cache hit rate	Second-request cache-read tokens / second-request prompt tokens	Higher is better
Observed cost	Sum of first + second request usage/cost values	Lower is better
Throughput	Completion tokens / request latency seconds	Higher is better
E2E latency	Full response latency per request	Lower is better
TTFT	Streaming first-token arrival time	Lower is better
Reasoning telemetry	Reasoning tokens from response usage	Used to explain latency, throughput, and cost

3. Experimental Environment and Data Quality

Item	Configuration
Model	`qwen/qwen3.7-flash`
Provider model IDs	infron: `qwen/qwen3.7-flash`; openrouter: `qwen/qwen3.7-flash`
Platforms	Infron and OpenRouter
API protocol	`/v1/chat/completions`
Routing modes	Throughput First, Price First, Latency First, TTFT First
Groups	4
Rounds per group	50
Workers	24
Request mode	Streaming Chat Completions with TTFT collection
Reasoning / thinking control	No explicit reasoning/thinking parameter; model and platform defaults are preserved
Prompt length tiers	`short` ≈1500, `medium` ≈8000, `long` ≈32000
Excluded records	230

4. Results: Overall Metrics and Main Findings

Routing mode	Platform	Strict pairs	Total Input Tokens	Token cache hit rate	Observed cost	Throughput	E2E latency	Streaming TTFT
Throughput First	Infron	163	5667484	53.89%	**$0.14534700**	**71.44 tok/s**	5906.04 ms	3006.71 ms
Throughput First	**OpenRouter**	163	5667484	**67.40%**	$0.20046654	5.97 tok/s	**3016.26 ms**	**2890.54 ms**
Price First	**Infron**	178	6497986	67.46%	**$0.12428800**	**69.26 tok/s**	6287.51 ms	**3247.79 ms**
Price First	OpenRouter	178	6497986	**70.11%**	$0.20870199	5.23 tok/s	**3440.89 ms**	3286.44 ms
Latency First	**Infron**	177	6086188	**69.29%**	**$0.11321900**	**73.55 tok/s**	6072.71 ms	3035.85 ms
Latency First	OpenRouter	177	6086188	69.13%	$0.19051714	5.85 tok/s	**3077.13 ms**	**2935.90 ms**
TTFT First	Infron	167	5711636	59.02%	**$0.11308200**	**68.65 tok/s**	6134.60 ms	3031.17 ms
TTFT First	**OpenRouter**	167	5711636	**61.90%**	$0.19892656	5.75 tok/s	**3127.46 ms**	**2988.47 ms**

4.1 Tail Latency and Statistical Tests

Tail percentiles expose risk that averages hide.

Routing mode	Platform	P50 latency	P95 latency	P99 latency	P50 TTFT	P95 TTFT	P99 TTFT
Throughput First	Infron	5699.36 ms	8905.14 ms	10907.52 ms	**2566.90 ms**	5920.58 ms	7224.64 ms
Throughput First	**OpenRouter**	**2837.99 ms**	**5214.80 ms**	**6625.83 ms**	2720.70 ms	**5109.37 ms**	**6027.52 ms**
Price First	**Infron**	6031.07 ms	9032.75 ms	12245.83 ms	**3066.21 ms**	**5959.19 ms**	**7922.48 ms**
Price First	OpenRouter	**3201.89 ms**	**6689.85 ms**	**8921.92 ms**	3076.23 ms	6177.10 ms	8866.79 ms
Latency First	**Infron**	5872.34 ms	8723.91 ms	9531.46 ms	**2841.24 ms**	5575.12 ms	**6260.39 ms**
Latency First	OpenRouter	**2969.81 ms**	**5456.74 ms**	**7410.76 ms**	2851.19 ms	**5340.58 ms**	7206.51 ms
TTFT First	Infron	5929.55 ms	8870.69 ms	10394.01 ms	2977.10 ms	**5491.02 ms**	**6103.65 ms**
TTFT First	**OpenRouter**	**3014.78 ms**	**6096.69 ms**	**8425.05 ms**	**2844.58 ms**	5551.33 ms	8139.81 ms

Mean deltas use bootstrap 95% CIs; p-values use paired sign-flip permutation tests.

Routing mode	Metric	Mean delta	95% CI	p-value	Pairs	Interpretation
Throughput First	Latency: OpenRouter – Infron	**–5779.54 ms**	–6086.35 ms to –5471.26 ms	<0.001	163	Positive means lower Infron latency
Throughput First	TTFT: OpenRouter – Infron	**–232.34 ms**	–453.01 ms to –0.10 ms	0.0500	163	Positive means lower Infron TTFT
Throughput First	Throughput: Infron – OpenRouter	**66.0393 tok/s**	63.6555 tok/s to 68.2486 tok/s	<0.001	163	Positive means higher Infron throughput
Throughput First	Cost: OpenRouter – Infron	**$0.00033816**	$0.00018829 to $0.00049834	<0.001	163	Positive means lower Infron cost
Throughput First	Token Cache Hit: Infron – OpenRouter	**–5.32 pp**	–12.32 pp to 1.80 pp	0.1492	163	Positive means higher Infron cache hit
Price First	Latency: OpenRouter – Infron	**–5693.25 ms**	–6166.21 ms to –5252.16 ms	<0.001	178	Positive means lower Infron latency
Price First	TTFT: OpenRouter – Infron	**77.30 ms**	–189.34 ms to 346.93 ms	0.5714	178	Positive means lower Infron TTFT
Price First	Throughput: Infron – OpenRouter	**63.7745 tok/s**	61.9940 tok/s to 65.4792 tok/s	<0.001	178	Positive means higher Infron throughput
Price First	Cost: OpenRouter – Infron	**$0.00047424**	$0.00031208 to $0.00064485	<0.001	178	Positive means lower Infron cost
Price First	Token Cache Hit: Infron – OpenRouter	**1.70 pp**	–4.45 pp to 8.44 pp	0.6383	178	Positive means higher Infron cache hit
Latency First	Latency: OpenRouter – Infron	**–5991.16 ms**	–6341.65 ms to –5632.86 ms	<0.001	177	Positive means lower Infron latency
Latency First	TTFT: OpenRouter – Infron	**–199.90 ms**	–412.01 ms to 14.55 ms	0.0650	177	Positive means lower Infron TTFT

Routing mode	Metric	Mean delta	95% CI	p–value	Pairs	Interpretation
Latency First	Throughput: Infron – OpenRouter	**67.6055 tok/s**	65.6004 tok/s to 69.3930 tok/s	<0.001	177	Positive means higher Infron throughput
Latency First	Cost: OpenRouter – Infron	**$0.00043671**	$0.00030013 to $0.00058454	<0.001	177	Positive means lower Infron cost
Latency First	Token Cache Hit: Infron – OpenRouter	**–1.34 pp**	–7.30 pp to 4.31 pp	0.7091	177	Positive means higher Infron cache hit
TTFT First	Latency: OpenRouter – Infron	**–6014.30 ms**	–6454.41 ms to –5574.72 ms	<0.001	167	Positive means lower Infron latency
TTFT First	TTFT: OpenRouter – Infron	**–85.40 ms**	–337.75 ms to 171.51 ms	0.5244	167	Positive means lower Infron TTFT
TTFT First	Throughput: Infron – OpenRouter	**62.7361 tok/s**	61.0764 tok/s to 64.4892 tok/s	<0.001	167	Positive means higher Infron throughput
TTFT First	Cost: OpenRouter – Infron	**$0.00051404**	$0.00033418 to $0.00070593	<0.001	167	Positive means lower Infron cost
TTFT First	Token Cache Hit: Infron – OpenRouter	**–3.07 pp**	–9.35 pp to 3.77 pp	0.4054	167	Positive means higher Infron cache hit

4.2 Reasoning / Thinking Control Check

This run does not explicitly set reasoning/thinking parameters and keeps model/platform defaults; this table records reasoning telemetry under default behavior.

Routing mode	Platform	Reasoning tokens	Avg reasoning tokens/request	Reasoning requests	Avg first reasoning token	Avg TTFT	Avg E2E latency
Throughput First	Infron	134940	413.9264	**326**	3036.10 ms	3006.71 ms	5906.04 ms
Throughput First	**OpenRouter**	**5216**	**16.0000**	**326**	**2890.54 ms**	**2890.54 ms**	**3016.26 ms**
Price First	**Infron**	152191	427.5028	**356**	**3274.56 ms**	**3247.79 ms**	6287.51 ms
Price First	OpenRouter	**5696**	**16.0000**	**356**	3286.44 ms	3286.44 ms	**3440.89 ms**
Latency First	**Infron**	155291	438.6751	**354**	3060.04 ms	3035.85 ms	6072.71 ms
Latency First	OpenRouter	**5664**	**16.0000**	**354**	**2935.90 ms**	**2935.90 ms**	**3077.13 ms**
TTFT First	Infron	137991	413.1467	**334**	3059.04 ms	3031.17 ms	6134.60 ms
TTFT First	**OpenRouter**	**5344**	**16.0000**	**334**	**2988.47 ms**	**2988.47 ms**	**3127.46 ms**

4.3 API Protocol Compatibility Matrix

This run uses `/v1/chat/completions`; this table records success response, usage, cost, and cache telemetry coverage for both platforms under that protocol.

API protocol	Endpoint	Platform	Pairs	Requests	Success rate	Usage coverage	Token usage coverage	Cost coverage	Cache telemetry coverage
`chat_completions`	`/v1/chat/completions`	Infron	800	1600	**100.00%**	**100.00%**	**100.00%**	**100.00%**	**100.00%**
`chat_completions`	`/v1/chat/completions`	OpenRouter	800	1600	92.81%	**100.00%**	**100.00%**	**100.00%**	**100.00%**

5. Result Visualizations by Routing Mode

The following charts are driven by the strict-paired summary data and present routing-mode, platform, cost, cache, E2E latency, Streaming TTFT, and upstream provider differences.

5.1 Core Metric Chart Overview

The charts below are generated from the same strict-paired summary data.

Figure 3-E2: Normalized capability contour

Five metrics are normalized to 0-100, where higher is better.

Figure 3-E3: Cost-cache efficiency plane

X-axis is token cache hit rate; Y-axis is observed total cost.

Figure 3-E1: Routing-mode core metric comparison

Infron and OpenRouter are shown side by side by routing mode.

Figure 3-E4: E2E latency and Streaming TTFT

Solid lines show E2E latency; dashed lines show Streaming TTFT.

Figure 3-E5: Upstream provider distribution

Provider shares explain cache-domain and performance differences.

Routing-Mode Drill-Down Charts

The horizontal axis shows relative advantage: blue to the right indicates Infron advantage, orange to the left indicates OpenRouter advantage.

Figure 4–E1: Throughput First core metric comparison

Direction and magnitude of relative advantage across five metrics.

Figure 4–E2: Price First core metric comparison

Direction and magnitude of relative advantage across five metrics.

Figure 4–E3: Latency First core metric comparison

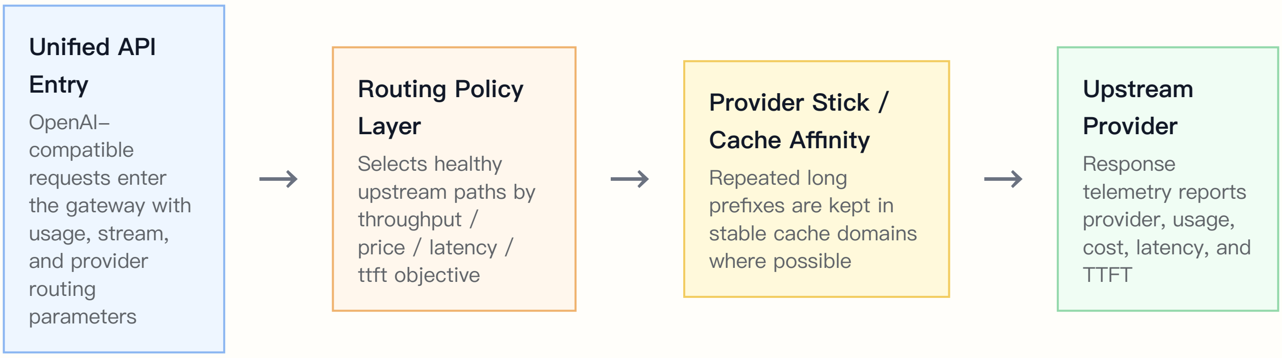
Direction and magnitude of relative advantage across five metrics.

Figure 4–E4: TTFT First core metric comparison

Direction and magnitude of relative advantage across five metrics.

6. Infron Technical Architecture and Cache/Cost Mechanism

Figure 12: Infron Multi-Provider Routing and Cache Control Plane



6.1 Multi-Provider Routing and Observable Control Plane

Requests enter a unified API, and the routing layer selects healthy upstream paths according to throughput, price, latency, or ttft objectives. Whether stable long prefixes stay in the same cache domain directly affects second-request cache-read tokens.

6.2 Provider Stick and Cache Affinity

Provider stick is cache affinity, not a permanent provider lock. Its goal is to reduce cache-domain fragmentation while respecting health and SLA constraints.

6.3 Cost-Control Path

Mechanism	Cache-rate effect	Cost effect	Observable signal
Stable prefix detection	Repeated prefixes are more likely to hit cache	Reduces repeated prefill cost	Payload SHA-256 and second-request cache-read tokens
Provider stick / cache affinity	Reduces cross-domain cache fragmentation	Avoids repeated cache warm-up	Provider distribution and token cache hit rate
Health checks and fallback	Protects availability while sometimes sacrificing cache	Reduces failure cost	HTTP status, provider distribution, tail latency
Cost-aware routing	Prefers lower-cost paths under constraints	Reduces total and per-round cost	Observed cost, cost breakdown coverage, cache-read tokens

7. Provider/Route Drill-Down

Routing mode	Platform	Total requests	Attributed requests	Provider distribution
Throughput First	Infron	326	326	`alibaba/jp` 326
Throughput First	OpenRouter	326	326	`Alibaba` 326
Price First	Infron	356	356	`alibaba/jp` 356

Routing mode	Platform	Total requests	Attributed requests	Provider distribution
Price First	OpenRouter	356	356	`Alibaba` 356
Latency First	Infron	354	354	`alibaba/jp` 354
Latency First	OpenRouter	354	354	`Alibaba` 354
TTFT First	Infron	334	334	`alibaba/jp` 334
TTFT First	OpenRouter	334	334	`Alibaba` 334

Upstream Provider Detail Distribution

Routing mode	Platform	Upstream provider	Requests	Share	first/ second	Covered rounds	Avg TTFT	Avg latency	Prompt tokens	Completion tokens	Rea tok
Throughput First	Infron	`alibaba/jp`	326	100.00%	163/163	163	3006.71 ms	5906.04 ms	5667484	137546	134
Throughput First	OpenRouter	`Alibaba`	326	100.00%	163/163	163	2890.54 ms	3016.26 ms	5667484	5868	521
Price First	Infron	`alibaba/jp`	356	100.00%	178/178	178	3247.79 ms	6287.51 ms	6497986	155037	152
Price First	OpenRouter	`Alibaba`	356	100.00%	178/178	178	3286.44 ms	3440.89 ms	6497986	6408	569
Latency First	Infron	`alibaba/jp`	354	100.00%	177/177	177	3035.85 ms	6072.71 ms	6086188	158119	155
Latency First	OpenRouter	`Alibaba`	354	100.00%	177/177	177	2935.90 ms	3077.13 ms	6086188	6372	560
TTFT First	Infron	`alibaba/jp`	334	100.00%	167/167	167	3031.17 ms	6134.60 ms	5711636	140661	137
TTFT First	OpenRouter	`Alibaba`	334	100.00%	167/167	167	2988.47 ms	3127.46 ms	5711636	6012	534

7.1 Cache-Rate and Cost Divergence Drill-Down

This table combines cache, cost, provider distribution, and reasoning telemetry to explain routing-mode differences.

Routing mode	Cache-hit delta	Infron cost multiple	Infron top path	OpenRouter top path	Reasoning token delta	Main attribution
Throughput First	-13.51 pp	**0.73x**	**`alibaba/jp` 100.00%**	**`Alibaba` 100.00%**	+129724	Cache and cost move in different directions; evaluate with speed metrics
Price First	-2.65 pp	**0.60x**	**`alibaba/jp` 100.00%**	**`Alibaba` 100.00%**	+146495	Cache and cost move in different directions; evaluate with speed metrics

Routing mode	Cache-hit delta	Infron cost multiple	Infron top path	OpenRouter top path	Reasoning token delta	Main attribution
Latency First	**+0.16 pp**	**0.59x**	**`alibaba/jp`** **100.00%**	`Alibaba` 100.00%	+149627	Infron leads both cache and cost
TTFT First	−2.88 pp	**0.57x**	**`alibaba/jp`** **100.00%**	**`Alibaba`** **100.00%**	+132647	Cache and cost move in different directions; evaluate with speed metrics

8. Stratified Results: Prompt-Length Cache Performance

This section aggregates second-request cache-read tokens, token-level cache hit rate, observed cost, E2E latency, and Streaming TTFT by prompt-length tier. Bold cells mark the advantaged side within each tier.

Prompt-Length Tier Overview

Prompt length tier	Target tokens	Platform	Pairs	Second prompt tokens	Second cache read tokens	Token cache hit rate	Observed cost	Avg E2E latency	Avg TTFT
`short`	1500	Infron	**231**	**460111**	**0**	**0.00%**	**$0.02354300**	5117.83 ms	**1788.12 ms**
`short`	1500	OpenRouter	**231**	**460111**	**0**	**0.00%**	$0.02868774	**2068.23 ms**	1915.67 ms
`medium`	8000	Infron	**232**	**2395822**	**697728**	**29.12%**	**$0.06336200**	6656.13 ms	**2889.94 ms**
`medium`	8000	OpenRouter	**232**	**2395822**	695680	29.04%	$0.11035802	**3182.88 ms**	3038.52 ms
`long`	32000	Infron	**222**	**9125714**	6815232	74.68%	**$0.40903100**	6553.02 ms	4631.66 ms
`long`	32000	OpenRouter	**222**	**9125714**	**7363584**	**80.69%**	$0.65956648	**4301.25 ms**	**4177.56 ms**

Prompt Length x Routing Mode Cache Hit Rate

Prompt length tier	Routing mode	Infron	OpenRouter	Winner
`short`	Throughput First	**0.00%**	**0.00%**	tie
`short`	Price First	**0.00%**	**0.00%**	tie
`short`	Latency First	**0.00%**	**0.00%**	tie
`short`	TTFT First	**0.00%**	**0.00%**	tie
`medium`	Throughput First	**46.51%**	45.01%	**Infron**
`medium`	Price First	**30.84%**	19.63%	**Infron**

Prompt length tier	Routing mode	Infron	OpenRouter	Winner
`medium`	Latency First	24.23%	**29.77%**	**OpenRouter**
`medium`	TTFT First	14.25%	**21.99%**	**OpenRouter**
`long`	Throughput First	58.64%	**76.96%**	**OpenRouter**
`long`	Price First	79.66%	**85.91%**	**OpenRouter**
`long`	Latency First	**85.07%**	83.37%	**Infron**
`long`	TTFT First	73.71%	**75.51%**	**OpenRouter**

9. Stratified Results: Group-Level Stability

Throughput First

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	1	48	48	52.96%	**$0.04762900**	8666.08 ms	5683.02 ms
Infron	2	30	30	45.17%	**$0.02461800**	7759.77 ms	6049.56 ms
Infron	3	44	44	57.45%	**$0.03744000**	9870.37 ms	6103.58 ms
Infron	4	41	41	57.93%	**$0.03566000**	8238.86 ms	5938.17 ms
OpenRouter	1	48	48	**67.37%**	$0.07207349	**4993.49 ms**	**4873.86 ms**
OpenRouter	2	30	30	**73.60%**	$0.03168526	**5560.44 ms**	**5493.56 ms**
OpenRouter	3	44	44	**57.96%**	$0.05789421	**5840.77 ms**	**5355.73 ms**
OpenRouter	4	41	41	**72.59%**	$0.03881359	**4670.82 ms**	**4629.94 ms**

Price First

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	1	50	50	**69.92%**	**$0.03242800**	8868.98 ms	5958.35 ms
Infron	2	37	37	**82.69%**	**$0.02072400**	8286.73 ms	**5139.67 ms**
Infron	3	49	49	53.45%	**$0.03863600**	9370.96 ms	**6172.76 ms**
Infron	4	42	42	67.35%	**$0.03250000**	10016.01 ms	**6200.28 ms**
OpenRouter	1	50	50	67.75%	$0.05904559	**5330.42 ms**	**5267.33 ms**
OpenRouter	2	37	37	63.12%	$0.04862474	**5749.84 ms**	5587.03 ms
OpenRouter	3	49	49	**71.49%**	$0.05522492	**7698.67 ms**	7418.68 ms
OpenRouter	4	42	42	**77.20%**	$0.04580675	**7145.65 ms**	6966.98 ms

Latency First

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	1	44	44	**71.92%**	**$0.02755200**	8759.05 ms	5893.91 ms
Infron	2	45	45	67.47%	**$0.03075500**	7965.10 ms	**5486.28 ms**
Infron	3	41	41	60.29%	**$0.02772500**	9171.96 ms	5379.96 ms
Infron	4	47	47	**76.84%**	**$0.02718700**	8400.30 ms	**4580.27 ms**
OpenRouter	1	44	44	56.18%	$0.04311926	**5234.62 ms**	**5153.55 ms**
OpenRouter	2	45	45	**71.13%**	$0.05412047	**5674.84 ms**	5609.34 ms
OpenRouter	3	41	41	**77.61%**	$0.04212614	**4978.07 ms**	**4414.54 ms**
OpenRouter	4	47	47	70.54%	$0.05115127	**4818.79 ms**	4746.44 ms

TTFT First

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	1	47	47	57.88%	**$0.03227800**	8318.36 ms	5404.08 ms
Infron	2	38	38	47.51%	**$0.02506900**	8721.57 ms	5370.84 ms
Infron	3	38	38	**66.97%**	**$0.02551800**	9881.11 ms	**5421.53 ms**
Infron	4	44	44	**62.19%**	**$0.03021700**	9091.01 ms	5708.10 ms
OpenRouter	1	47	47	**62.75%**	$0.05790167	**5056.57 ms**	**4933.56 ms**
OpenRouter	2	38	38	**77.32%**	$0.03294962	**4261.46 ms**	**4207.43 ms**
OpenRouter	3	38	38	54.10%	$0.05359975	**7012.86 ms**	6867.62 ms
OpenRouter	4	44	44	55.54%	$0.05447553	**5279.44 ms**	**5155.22 ms**

10. Discussion: Business Value, Boundaries, and Engineering Implications

Business decisions should not rely on one metric. Stable long-context and high-frequency template workloads should prioritize cache rate and cost; realtime interaction must constrain TTFT and E2E latency; batch processing often prioritizes throughput and failure cost.

Routing mode	Business objective	Observed result	Scenarios	Caveat
Throughput First	Maximize output capacity per unit time	OpenRouter leads overall (3/5 metrics)	Batch generation, offline summaries, backend processing	First-token and E2E response are faster; check cache and cost
Price First	Minimize request and token cost	Infron leads overall (3/5 metrics)	High-frequency templates, support automation, marketing, RAG prefixes	Decide with budget, SLA, and cache stability together

Routing mode	Business objective	Observed result	Scenarios	Caveat
Latency First	Minimize full-response waiting time	Infron leads overall (3/5 metrics)	Online chat, agent chains, IDE/writing assistants, realtime tools	Cache and cost are stronger; still check speed SLA
TTFT First	Minimize streaming first-token time	OpenRouter leads overall (3/5 metrics)	Streaming chat, realtime copilots, first-screen feedback	First-token and E2E response are faster; check cache and cost

11. Conclusion

Routing-Mode Conclusions

Routing mode	Objective	Cache winner	Cost winner	Throughput winner	E2E latency winner	TTFT winner	Interpretation
Throughput First	Maximize output capacity per unit time	**OpenRouter**	**Infron**	**Infron**	**OpenRouter**	**OpenRouter**	OpenRouter leads overall (3/5 metrics)
Price First	Minimize request and token cost	**OpenRouter**	**Infron**	**Infron**	**OpenRouter**	**Infron**	Infron leads overall (3/5 metrics)
Latency First	Minimize full-response waiting time	**Infron**	**Infron**	**Infron**	**OpenRouter**	**OpenRouter**	Infron leads overall (3/5 metrics)
TTFT First	Minimize streaming first-token time	**OpenRouter**	**Infron**	**Infron**	**OpenRouter**	**OpenRouter**	OpenRouter leads overall (3/5 metrics)

12. Limitations, Missing Data, and Next Experiments

Limitation	Impact	Next step	Current handling
Full routing trace	Cannot prove every provider choice, fallback, and retry path hop by hop	Add provider routing trace, decision logs, and fallback reasons	Use only returned provider fields and provider distribution
Longer time window	4x50 observes short-window stability but not day-level drift	Add soak tests and repeated windows	Scope conclusions to this run
Production corpus	Built-in templates do not cover every workload distribution	Use sanitized production-stratified corpora	Discuss representative long-context templates only
Cost-field consistency	Cost coverage and semantics may differ by platform	Reconcile with billing and provider cost breakdown	Use only explicitly returned cost fields

13. Reproducibility Appendix

Artifact	Path
Summary	`export/qwen3_7_flash_all_experiments/` `routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions` `summary.json`
Paired dataset	`export/qwen3_7_flash_all_experiments/` `routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions` `benchmark_pairs.csv`
Request-level dataset	`export/qwen3_7_flash_all_experiments/` `routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions` `benchmark_requests.jsonl`
Filtered structured records	`export/qwen3_7_flash_all_experiments/` `routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions` `records.json`
Excluded-record audit	`export/qwen3_7_flash_all_experiments/` `routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions` `records_excluded.json`
Test source	`tests/test_rerun_routing_sort_cache_cost_ab.py`
Benchmark runner source	`scripts/rerun_routing_sort_cache_cost_ab.py`
HTML report renderer source	`scripts/render_glm52_deepseek_style_report.py`
Dataset reference	`business_representative` built-in representative business templates; request-level export is `benchmark_requests`